

Cell Injury: Overview

Cell injury is the result of stress that exceeds the adaptive capacity of the cell.

Hypoxia is the most common cause of cell injury in clinical practice.

Reversible injury is characterised by cellular swelling and fatty change.

The ATP Depletion Cascade

ATP depletion is the central event that follows a sustained fall in oxygen delivery.

The sodium potassium pump consumes a large share of the cellular ATP budget.

Pump failure causes intracellular sodium and water to accumulate inside the cell.

Morphology of Reversible Injury

Cellular swelling is the earliest morphological change of reversible cell injury.

Hydropic change refers to the accumulation of water within the injured cell.

Membrane blebbing occurs when the cytoskeleton detaches from the plasma membrane.

Irreversible Injury

Severe membrane damage marks the transition to irreversible cell injury.

Mitochondrial dysfunction leads to the permanent loss of ATP generation.

Necrosis is the pattern of cell death that follows irreversible injury in living tissue.